Final Project
Marketing Campaign Optimization

Table of Contents

1. Project Objective **2**
2. Dataset **2**
3. Business Scenario **3**
4. Problem Formulation **3**
Part A: Causal Inference Analysis 3
Part B: Uplift Modeling 3
5. Common Pipeline **4**
5.1 Data Preprocessing 4
5.2 Exploratory Data Analysis 4
5.3 Causal Inference Analysis 5
5.4 Uplift Modeling 5
5.5 Campaign Targeting Policy 5
6. Causal Inference Analysis **6**
6.1 Treatment Assignment Analysis 6
6.2 Causal Graph (DAG) 6
6.3 Observable vs. Unobservable Confounding 6
6.4 Estimator Comparison 7
6.5 Business Interpretation 7
7. Evaluation **8**
7.1 Causal Estimation 8
7.2 Uplift Modeling 8
7.3 Policy Evaluation 8
7.4 Ablation Study 8
8. Engineering Requirements **9**
Suggested Project Structure 9
Technical Requirements 10
Code Quality 10
9. Expected Outcome **10**

1. Project Objective

The goal of this project is to build an end-to-end decision support system for **Marketing Campaign Optimization** using **Causal Inference** and **Uplift Modeling**.

Instead of sending promotional campaigns to every customer, the objective is to estimate:

- The average impact of a campaign (ATE)
- The heterogeneous treatment effect for each customer (CATE)
- Which customers should receive the campaign
- Which customers should not receive any treatment
- The expected business impact under different targeting policies

Students will implement a complete pipeline from raw data to causal analysis, uplift modeling, policy optimization, and business evaluation.

The project should include:

- Data preprocessing
- Exploratory data analysis
- Causal assumption analysis
- Average Treatment Effect estimation
- Individual Treatment Effect estimation
- Uplift modeling
- Campaign targeting policy
- Offline evaluation
- Business impact analysis

2. Dataset

Dataset Used: Criteo Uplift Prediction Dataset (https://ailab.criteo.com/criteo-uplift-prediction-dataset/)

The dataset contains customer-level information collected during a digital advertising campaign.

Each observation includes:

- Customer features
- Treatment assignment
- Conversion outcome

The dataset is designed for causal inference and uplift modeling research.

3. Business Scenario

A marketing team plans to launch a promotional campaign.

However, contacting every customer is expensive and may even reduce customer engagement if promotions are sent to the wrong audience.

The business wants to answer the following questions:

- Does the campaign actually increase conversions?
- What is the overall campaign effectiveness?
- Which customers benefit from the campaign?
- Which customers should not receive the campaign?
- What targeting strategy maximizes business value?

The final objective is to maximize incremental conversions while minimizing campaign cost.

4. Problem Formulation

The project consists of two major components.

Part A: Causal Inference Analysis

Estimate the causal effect of the marketing campaign.

Students should estimate:

- Average Treatment Effect (ATE)
- Conditional Average Treatment Effect (CATE)

Supported estimation methods include:

- Difference in Means
- Regression Adjustment
- Propensity Score Matching
- Inverse Propensity Weighting (IPW)
- Doubly Robust Estimation (DR)

Students should compare different estimators and discuss their assumptions.

Part B: Uplift Modeling

Build models to estimate individual treatment effects.

Supported approaches include:

Meta Learners

- S-Learner
- T-Learner
- X-Learner
- R-Learner
- DR-Learner

Tree-based Methods

- Causal Tree
- Causal Forest
- Uplift Tree

Representation-based Methods

- DragonNet

The output should be:

- Estimated CATE
- Customer uplift score
- Customer ranking
- Campaign recommendation

5. Common Pipeline

5.1 Data Preprocessing

Perform data cleaning.

Examples include:

- Handle missing values
- Remove invalid records
- Normalize numerical features (if necessary)

5.2 Exploratory Data Analysis

Perform exploratory analysis to understand:

- Treatment ratio

- Conversion rate
- Feature distributions
- Missing values
- Feature correlations

Students should discuss potential biases observed in the dataset.

5.3 Causal Inference Analysis

Estimate the overall treatment effect using multiple causal estimators.

Compare:

- Difference in Means
- Regression Adjustment
- IPW
- Doubly Robust

Discuss:

- estimation differences
- robustness
- assumptions

5.4 Uplift Modeling

Train one or more uplift models in Section 4.

5.5 Campaign Targeting Policy

Design a marketing strategy based on estimated uplift.

Possible targeting strategies include:

- Treat everyone
- Treat nobody
- Random targeting
- Treat customers with positive uplift
- Treat Top 10% uplift customers
- Treat Top 20% uplift customers
- Treat Top-K customers maximizing expected business gain

Compare the effectiveness of different targeting policies.

6. Causal Inference Analysis

Students must provide a complete causal analysis instead of only training machine learning models.

6.1 Treatment Assignment Analysis

Investigate whether the treatment assignment appears randomized.

Possible analyses include:

- Treatment distribution
- Covariate balance
- Standardized Mean Difference (SMD)
- Logistic regression predicting treatment
- Propensity score distribution

Questions to discuss:

- Does treatment appear randomized ?
- Can customer features predict treatment assignment ?
- What evidence supports your conclusion ?

6.2 Causal Graph (DAG)

Construct a causal graph representing the business scenario.

The graph should identify:

- Treatment
- Outcome
- Potential confounders
- Possible mediators
- Possible colliders

Discuss whether the causal effect is identifiable.

6.3 Observable vs. Unobservable Confounding

Assume two scenarios.

Scenario 1

Treatment assignment is randomized.

Discuss:

- Why simple Difference in Means consistently estimates ATE.
- Whether propensity score adjustment is still necessary.

Scenario 2

Treatment assignment is observational.

Discuss:

- Possible confounders.
- Why naive estimation becomes biased.
- Which adjustment methods should be applied.
- Which assumptions become necessary.

6.4 Estimator Comparison

Estimate ATE using multiple causal estimators.

Compare:

- Estimated ATE
- Bias and variance
- Computational complexity
- Practical advantages and disadvantages

Discuss when each estimator should be preferred.

6.5 Business Interpretation

Interpret the estimated treatment effect.

Examples:

- Is the campaign effective ?
- Should the campaign continue ?
- Which customer segments benefit the most ?
- Which segments may be harmed by the treatment ?

Support conclusions using both numerical results and visualizations.

7. Evaluation

7.1 Causal Estimation

Evaluate:

- Estimated ATE
- ATT (optional)
- CATE distribution
- Confidence intervals

7.2 Uplift Modeling

Evaluate uplift performance using:

- Qini Curve
- AUUC
- Uplift Curve
- Incremental Gain Curve
- Calibration by Decile
- Average Uplift

Students should compare multiple uplift models.

7.3 Policy Evaluation

Compare different targeting policies.

Possible metrics include:

- Incremental conversions
- Incremental gain
- Campaign coverage

Discuss the trade-off between campaign size and business value.

7.4 Ablation Study

Compare different estimation approaches.

Examples:

- Difference in Means vs IPW
- IPW vs Doubly Robust
- S-Learner vs T-Learner
- T-Learner vs X-Learner
- X-Learner vs DR-Learner
- Meta Learners vs Causal Forest

Discuss why performance differs.

8. Engineering Requirements

Suggested Project Structure

project/

├── data/

├── notebooks/

├── features/

├── causal/

│ ├── ate/

│ ├── propensity/

│ └── doubly_robust/

├── uplift/

│ ├── meta_learners/

│ ├── tree_models/

│ └── deep_models/

├── evaluation/

├── visualization/

├── report/

└── README.md

Technical Requirements

Recommended libraries:

- Python
- Pandas
- NumPy
- Scikit-learn
- EconML
- CausalML
- DoWhy (optional)
- Matplotlib
- XGBoost / LightGBM / CatBoost

Code Quality

The project should provide:

- Reproducible experiments
- Modular implementation
- Well-documented code
- Clear visualization
- GitHub repository with README

9. Expected Outcome

By completing this project, students will be able to:

- Formulate business problems using the potential outcomes framework.
- Analyze treatment assignment mechanisms and evaluate whether causal identification assumptions are plausible.
- Estimate Average Treatment Effects (ATE) using multiple causal inference methods.
- Estimate Conditional Average Treatment Effects (CATE) with modern uplift modeling techniques.
- Compare meta-learners, tree-based methods, and representation-based models for heterogeneous treatment effect estimation.
- Design data-driven campaign targeting policies based on predicted uplift rather than response prediction alone.
- Evaluate targeting strategies using uplift-specific metrics such as Qini Curve, AUUC, and Incremental Gain.
- Translate causal inference results into actionable business recommendations for marketing campaign optimization.

- Build an end-to-end causal analytics workflow that integrates data preprocessing, causal estimation, uplift modeling, policy optimization, and business evaluation.